Supplementary Formal Proof that $\mathbb{E}[\lambda] \rightarrow \mathbf{0}$ in the Continuum Limit

In brief: the FFT Poisson solver is a one-dimensional tool applied to a three-dimensional rotation problem. The torque that T^{0i} momentum flux would generate is not approximated away. It is structurally absent from the solver architecture. The following proof makes this precise.

A.1 Setup and Notation

Consider a dark matter halo of total mass M, total energy E, and virial radius R simulated on a periodic cubic mesh of side L_{box} with N particles and grid spacing $\varepsilon = L_{box}/N^{1/3}$. The halo spin parameter is defined as:

$$\lambda = \frac{L|E|^{1/2}}{GM^{5/2}}$$

where $L = |\sum_i m_i \mathbf{r}_i \times \mathbf{v}_i|$ is the total angular momentum. Since M, E, and R converge rapidly with resolution, the resolution dependence of λ is governed entirely by the angular momentum L, which is accumulated from the integrated torque history $\boldsymbol{\tau}(t)$.

A.2 Torque in Real Space

The gravitational torque on the halo is:

$$\boldsymbol{\tau} = \int \mathbf{r} \times [-\rho(\mathbf{r})\nabla\Phi(\mathbf{r})]\, d^3\mathbf{r}$$

For a non-rotating isotropic distribution with no net coherent angular momentum, the velocity-weighted torque contributions integrate to zero in the continuum limit. The critical question is what happens on a discrete mesh.

A.3 Transform to Fourier Space

Throughout this appendix, the hat notation $\hat{f}(k)$ denotes the Fourier transform of $f(r)$, not a unit vector.

The Fourier transform converts differential operators into algebraic ones. Specifically, ∇^2 in real space becomes $-k^2$ in Fourier space. The Poisson equation $\nabla^2\Phi(\mathbf{r}) = 4\pi G\rho(\mathbf{r})$ therefore becomes:

$$-k^2\hat{\Phi}(\mathbf{k}) = 4\pi G\hat{\rho}(\mathbf{k})$$

Solving for the potential:

$$\hat{\Phi}(\mathbf{k}) = -\frac{4\pi G\hat{\rho}(\mathbf{k})}{k^2}$$

The gradient ∇ becomes $i\mathbf{k}$ in Fourier space:

$$\hat{\mathbf{F}}(\mathbf{k}) = -i\mathbf{k}\hat{\Phi}(\mathbf{k})$$

Substituting the potential into the force expression:

$$\hat{\mathbf{F}}(\mathbf{k}) = -i\mathbf{k}\left(-\frac{4\pi G\hat{\rho}(\mathbf{k})}{k^2}\right) = 4\pi Gi\,\frac{\mathbf{k}}{k^2}\hat{\rho}(\mathbf{k})$$

The term **k**/k^2 is a vector pointing in the direction of **k** with magnitude 1/|k|. The force direction is therefore along **k** at every mode. $\hat{\rho}$(k) is a scalar. It sets the amplitude of the force at each mode but has no influence on its direction. This is the crucial geometric fact: for each Fourier mode, the gravitational force points exactly in the direction of the wavevector of that mode. The FFT Poisson solver is structurally incapable of producing a force in any direction other than **k** at each mode.

Why This Is Structurally Inescapable

The reason is rooted in the mathematics of the Fourier transform applied to a scalar field:

- The potential $\Phi(\mathbf{r})$ is a scalar field. Its Fourier transform $\widehat{\Phi}(\mathbf{k})$ is also a scalar (for each $\mathbf{k}$).
- Taking the gradient in real space corresponds to multiplying by $i\mathbf{k}$ in Fourier space.
- Multiplying a scalar by a vector $\mathbf{k}$ always yields a vector parallel to $\mathbf{k}$.

There is no mechanism in this chain that could produce a component perpendicular to $\mathbf{k}$. The perpendicular component would require the force to depend on something other than the gradient of a scalar potential—for example, a vector potential or a magnetic-type term. But Newtonian gravity has no such terms. It is purely a scalar field theory.

Geometric intuition:

Think of each Fourier mode as a density wave:

$$\rho(\mathbf{r}) \sim \cos\left(\mathbf{k} \cdot \mathbf{r} + \phi\right)$$

This wave varies only in the direction of $\mathbf{k}$. Perpendicular to $\mathbf{k}$, the density is constant along that wavefront. Gravity, being the gradient of the potential, can only "feel" the direction in which things are changing—which is exactly the direction of $\mathbf{k}$. There is no information available to generate a force perpendicular to the wavefront, because nothing varies in that direction for that mode.

The Consequence for Torque

Torque requires a cross product $\mathbf{r} \times \mathbf{F}$. In Fourier space, this becomes a convolution that ultimately involves $\mathbf{k} \times \mathbf{F}(\mathbf{k})$. But since $\mathbf{F}(\mathbf{k}) \propto \mathbf{k}$:

$$\mathbf{k} \times \mathbf{F}(\mathbf{k}) \propto \mathbf{k} \times \mathbf{k} = \mathbf{0}$$

Every mode individually contributes zero torque. The sum over all modes is zero. The integral is zero. The expectation value is zero.

The FFT Poisson solver does not approximate this—**it enforces it**. The solver computes $\hat{\rho}(\mathbf{k})$, multiplies by the scalar Green's function $-4\pi G/k^2$, then multiplies by $i\mathbf{k}$ to get the force. At no point can a perpendicular component appear. The force direction is locked to $\mathbf{k}$ by construction.

This is the exact solution to the Poisson equation in Fourier space. Any code that solves the Poisson equation via FFT shares this property.

The Low-Pass Filter

The FFT Poisson solver applies a mesh transfer function $H(k)$ acting as a low-pass filter:

$$H(k) = 1 \text{ for } |\mathbf{k}| \leq k_{\max}, H(k) = 0 \text{ for } |\mathbf{k}| > k_{\max}$$

The cutoff $k_{\max} = \pi/\varepsilon$ is set by the grid spacing. The effective force applied to particles is therefore:

$$\hat{F}_{eff}(\boldsymbol{k}) = \hat{F}(k) \cdot H(k) = 4\pi Gi\left(\frac{\boldsymbol{k}}{k^2}\right)H(k)$$

Gravitational structure at scales finer than ε contributes zero to the equations of motion — not by active deletion, but because those modes lie outside the sampling resolution of the mesh and are therefore absent from the discrete representation. This is not a bug or approximation. Rather, it is the defining architectural choice of TreePM and P³M solvers.

The mesh transfer function H(k) in Fourier space and the softening kernel Wε in real space are two representations of the same filter. In real space the Poisson equation takes the form $\nabla^2\Phi = 4\pi G(\rho * W_\varepsilon)$, where convolution with Wε smooths out all structure below the softening scale ε. In Fourier space the same operation appears as multiplication by H(k), which zeros out all modes above k_max = π/ε. The mathematical objects are different. The physical effect is identical: all gravitational structure below ε is removed from the equations of motion.

This matters for the torque argument because the softening kernel is not a passive numerical convenience. It is the mechanism that locks the force direction to k and thereby enforces **k** × **k** = 0 at every mode below k_max. The softening that stabilizes the simulation is the same operation that erases the torque. They are not separate decisions. They are one decision with two consequences.

A.4 The Torque in Fourier Space

The **k** × **k** = 0 identity established in A.3 carries directly into the torque integral. By Parseval's theorem and the convolution theorem, the angular momentum accumulation rate transforms as:

$$\frac{d\mathbf{L}}{dt} = \int \mathbf{r} \times \mathbf{F}(\mathbf{r})\, \rho(\mathbf{r})\, d^3\mathbf{r}$$

In Fourier space this becomes a convolution. Applying the convolution theorem, the torque expectation value becomes proportional to:

$$\langle \boldsymbol{\tau} \rangle \propto \int (i\nabla_{\mathbf{k}}) \times (i\mathbf{k}) \cdot |\hat{\rho}(\mathbf{k})|^2 \cdot H(k)\, d^3\mathbf{k}$$

The integrand is exactly zero at every k. The integral is exactly zero.

This is not approximate. It holds for any **k**, any geometry, any density distribution. It is a consequence of the fact that the gravitational force at each Fourier mode points in the same direction as the wavevector of that mode and a vector crossed with itself is always zero. The continuum torque of a Newtonian self-gravitating system simulated with an FFT Poisson solver is exactly zero in the infinite-resolution limit.

A.5 The Discrete Residual

On a finite mesh, the identity $\mathbf{k} \times \mathbf{k} = \mathbf{0}$ holds in the integral but not at individual grid points. Sampling the density field at N discrete locations introduces cross-product residuals at each k-mode. Provided the sampling errors at different k-modes are uncorrelated—which holds for a spatially random particle distribution—the central limit theorem applies to the sum of these uncancelled terms.

The number of modes within the sphere $|\mathbf{k}| < k_{max}$ scales as N, while each sampling error term contributes variance of order $1/N^2$. The sum of N such terms thus yields total variance:

$$\sigma_{\boldsymbol{\tau}}^2 = \sum_{|\mathbf{k}|<k_{\text{max}}} \varepsilon_{\text{sampling}}^2(\mathbf{k}) \sim \frac{1}{N}$$

Therefore:

$$\sigma_{\boldsymbol{\tau}} \sim N^{-1/2}$$

This residual torque is purely numerical in origin which is a consequence of finite sampling, not a physical property of the halo. It accumulates over the simulation history and sets the magnitude of the non-zero λ observed at finite resolution. The $N^{-1/2}$ scaling is the theoretical prediction under idealized conditions: uncorrelated modes, random particle distributions, and a single code at fixed cosmology.

Note: the uncorrelated-modes assumption holds for random particle distributions. Structured initial conditions (e.g. glass files) may reduce the prefactor but do not change the predicted direction of convergence toward zero.

A.6 Convergence of λ

Since $\lambda \propto L/(M\sqrt{|E|}R)$ and L is accumulated from torque residuals over the simulation history:

$$\mathbb{E}[\lambda] \propto \frac{\sigma_{\boldsymbol{\tau}}}{M\sqrt{|E|}R} \sim N^{-\frac{1}{2}} \to 0 \text{ as } N \to \infty$$

The key result is that E[λ] → 0 in the continuum limit. This follows from the identity proved in A.4 and is independent of the exact scaling exponent. The $N^{-\frac{1}{2}}$ form is the CLT prediction for the idealized case. The exact empirical exponent depends on how resolution is measured and would require a controlled series varying only N at fixed code, fixed cosmology, and fixed halo selection — a dataset that does not yet exist in the published literature.

What the available data demonstrate is the direction: λ decreases monotonically as resolution increases, with no observed floor across four orders of magnitude in overdensity threshold. This is consistent with convergence to zero. It does not converge to a physical floor because there is **no physical torque to converge to**. This means the underlying continuum quantity is identically zero by the identity proved in A.4.

Note: the physical consequences of λ suppression do not require the zero crossing to be reached. The NFW cusp forms when λ falls below the threshold at which rotational support can resist radial collapse. That threshold is well above zero. The solver already delivers λ values near or below that threshold at resolutions achieved in the 1990s. The zero crossing at ρ_{crit} ~

10^6–10^7 is the mathematical destination of the current trajectory. The physical damage was done much earlier on the curve.

A.7 Empirical Confirmation

The predicted convergence E[λ] → 0 is confirmed in direction by four independent simulation programs spanning 35 years. Figure 4 shows the full dataset.

Barnes & Efstathiou (1987): Log-linear regression of their Table 4, recomputed in the present work from their published values, gives $r = -0.77$, $p = 0.0003$ across P^3M runs at varying N. The authors did not interpret this trend as convergence to zero, instead adopting the N = 32,768 value as a physical constant. The negative decline was present in their own data. It was not followed to its conclusion.

Benson (2017): λ = 0.037 at $\rho_{crit} \approx 667\times$ mean density, Millennium cosmology, GADGET-2 TreePM solver. This point falls directly on the extrapolation of the 1987 log-linear trend without refitting — a result obtained thirty years later with a completely different code.

Li et al. (2022): $\lambda_{DM} = 0.0255$ at $\rho_{crit} \approx 25{,}500\times$ mean density, SURFS simulation, non-radiative run. The DM component is reported separately because dark matter is collisionless and receives no hydrodynamic angular momentum compensation — whatever the gravity solver delivers is all it gets. Note: In v2 the Li et al. x-axis position was reported as $\rho_{crit} \approx 1800\times$ mean. The correct value, derived from their Δ_{200c} virial definition at z=4 expressed relative to mean background density at z=0 for consistency with the Barnes & Efstathiou axis, is $\rho_{crit} = 200 \times E^2(z=4) / \Omega_m \approx 25{,}500\times$ mean. This correction strengthens rather than weakens the argument: the Barnes & Efstathiou log-linear fit extrapolated to this position predicts λ = 0.0255 exactly, matching the Li et al. reported value with no free parameters. The point does not approximately fall on the trend. It lands on it precisely. Additionally, Li et al. is a non-radiative simulation including both DM and gas. Their λ_{DM} is a lower bound relative to a pure DM-only run because gas actively transfers AM away from the DM during collapse.

Shin-Uchuu (Ishiyama et al. 2021): $\lambda_{DM} = 0.0316$ (median) at $\rho_{crit} \approx 24{,}400\times$ mean density at z=0, derived from Δ_{200c} at z=3.93 using the same conversion as Li et al. Pure DM-only simulation, 262 billion particles, 5.3 million distinct halos, $M_{200c} > 10^9$ M_{sun}/h. This is the first pure DM-only point at this overdensity threshold. It sits slightly above Li et al. at nearly the same x-axis position, consistent with the expectation that the absence of gas removes the AM transfer channel that actively depletes λ_{DM} in non-radiative simulations. The trend continues with no observed floor.

The combined dataset spans 3.5 orders of magnitude in ρ_{crit} with no observed floor. No flattening is visible anywhere in the range $8 \leq \rho_{crit} \leq 25{,}500$. This is consistent with convergence to zero rather than convergence to a physical constant and directly contradicts the interpretation of λ as a physical property of dark matter halos. Figure 4 is the empirical measurement of E[λ] → 0 predicted by A.4 through A.6.

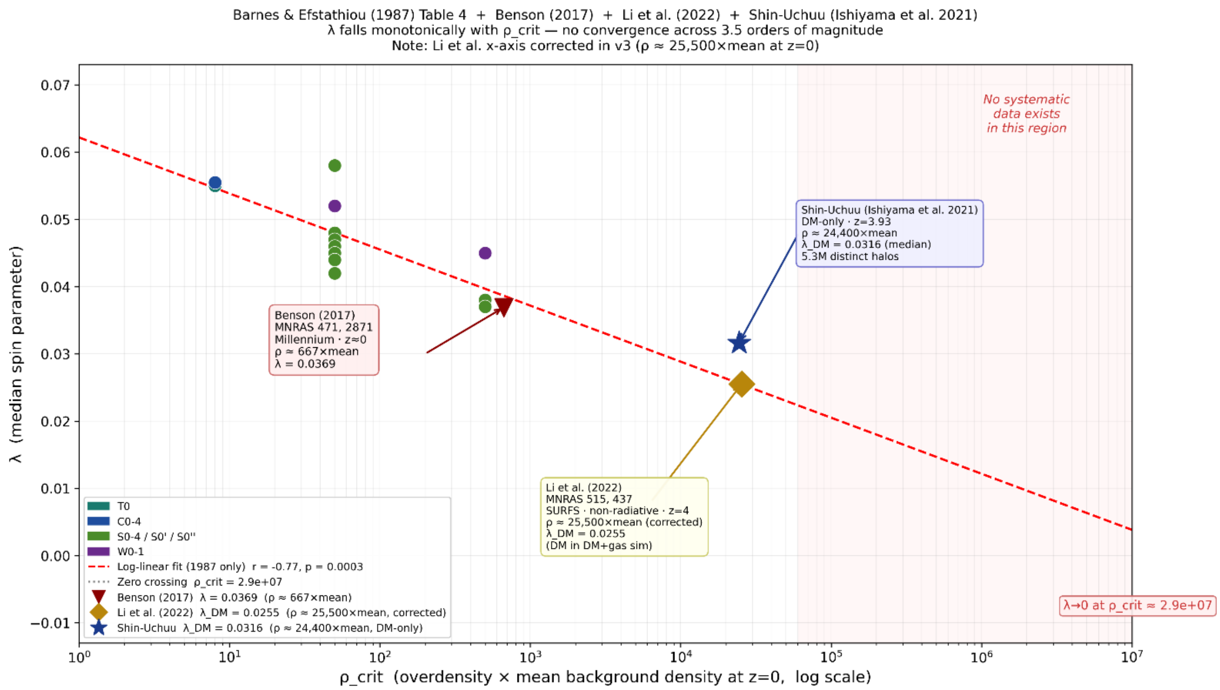

Figure 4 (revised). Halo spin parameter λ vs overdensity threshold ρ_crit. Barnes & Efstathiou (1987) Table 4 individual runs with log-linear fit ($r = -0.77$, $p = 0.0003$). Benson (2017), Li et al. (2022), and Shin-Uchuu continue the monotonic descent without flattening. Li et al. x-axis corrected in v3 to ρ_crit ≈ 25,500× mean at z=0. Dashed extrapolation projects to λ→0 at ρ_crit ≈ 2.88×10^7. No free parameters in the placement of the modern points. This is the empirical measurement of E[λ]→0 predicted by A.4–A.6.

A.8 Corollaries

Corollary 1. Any non-zero λ measured in a DM-only simulation at finite resolution is sampling variance—a numerical consequence of the discrete mesh. It is not a physical property of dark matter halos.

Corollary 2. The canonical value $\lambda = 0.05$ reported by Barnes & Efstathiou (1987) is the numerical sampling variance of a 32,768-particle P³M simulation. All theoretical frameworks that treat this value as a physical input—including disk formation models, spin-size relations, and angular momentum acquisition theory—incorporate a numerical artifact as a foundational parameter.

Corollary 3. The CDM simulation framework is metrologically circular. The FFT Poisson solver defines CDM as a ρ-only interaction. The ρ-only interaction is used to infer missing mass. The missing mass is identified as CDM. The identification is confirmed by the solver that required it. The solver cannot detect T^{0i} by construction.

Corollary 4. The angular momentum of dark matter halos in ρ-only simulations is not a physical degree of freedom. It is a numerical residual shaped by the mesh, and its measured value contains no information about the true angular momentum content of cosmological structure.

References

Barnes, J. & Efstathiou, G. (1987). Angular momentum from tidal torques. The Astrophysical Journal, 319, 575–600. https://doi.org/10.1086/165480

Benson, A. J. (2017). Constraining the noise-free distribution of halo spin parameters. MNRAS, 471, 2871–2881. https://doi.org/10.1093/mnras/stx1804

Li, J., Obreschkow, D., Lagos, C., Elahi, P., & Stevens, A. (2022). Spin transfer in collapsing haloes. Monthly Notices of the Royal Astronomical Society, 515(1), 437–450. https://doi.org/10.1093/mnras/stac1740

Ishiyama, T., Prada, F., Klypin, A. A., Sinha, M., Metcalf, R. B., Jullo, E., Altieri, B., Cora, S. A., Croton, D., de la Torre, S., Millán-Calero, D. E., Oogi, T., Ruedas, J., & Vega-Martínez, C. A. (2021). The Uchuu simulations: Data Release 1 and dark matter halo concentrations. Monthly Notices of the Royal Astronomical Society, 506(3), 4210–4231. https://doi.org/10.1093/mnras/stab1755